

A Practical Internet-Reachability Test for FedRAMP

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A plain-language companion to “Internet Reachability at Scale under the FedRAMP VDR/VER Rules.”

The short version

FedRAMP asks providers to decide, for every vulnerability finding, whether a payload from the public internet might trigger it. That is broader than asking whether the affected server has a public IP. A crafted request can pass through an application and reach a private database, parser, queue worker, or logging library.

The definition is technically sound. The operational problem is that ordinary scanner findings usually do not say whether a CVE can be triggered after its input crosses a component boundary. At scale, a provider cannot manually research that question for every finding on every asset.

The paper recommends a current triage rule:

THE WHOLE MODEL

A finding is internet-reachable when its vulnerable network service can be triggered directly, or when a known indirect trigger reaches the component and current assessment and continuous-monitoring evidence does not show that the relevant trigger class is prevented before vulnerable processing. An active AV:L finding is also promoted when the same verified execution boundary contains an active, directly reachable, high-confidence entry point that can plausibly supply its local-execution prerequisite.

That rule is intentionally practical. It uses information providers already have, promotes exceptional indirect triggers and execution-capable entry points through reusable CVE-level enrichment, and keeps a more detailed application-wide exploit-path model optional. A chained promotion changes only the finding’s deployment-specific IRV status: its AV:L vector remains AV:L, and the record names the upstream CVE that caused the promotion.

Why the scanner cannot finish the job

A scanner can usually identify the affected package, CVE, version, and CVSS vector. It may also report the listening service and a short vulnerability description. That is enough to automate much of the direct-exposure question.

The missing fact is the trigger path. A local parser flaw may be completely unrelated to internet input, or it may process every uploaded document. A logging library may have no listener at all but still receive attacker-controlled strings from a public API. CVSS attack vector helps describe the

vulnerable component; it does not reliably say whether another component can carry the trigger to it.

This is primarily an upstream metadata problem. Vendors, CNAs, vulnerability publishers, and scanner feeds should distribute a reusable field such as `indirect_internet_trigger`, with provenance and enough trigger detail to support review. They should also publish whether exploitation establishes local code or command execution, the supporting evidence, and the server, client, or other context in which that execution occurs. Today they generally do not.

Four checks, in order

The operational method asks four questions.

1. **Can the internet directly deliver what this flaw requires?** Match public routing, live listeners, protocol, version, and relevant edge controls to the vulnerability's direct trigger. A public host does not make every installed package reachable.
2. **Is this a known indirect-trigger vulnerability?** Record a reusable CVE-level status: confirmed, ruled out, or unknown. A vendor advisory, threat-intelligence report, exploit analysis, reviewed agentic proposal, or analyst decision can confirm the status once and reuse it across the fleet.
3. **Does the component receive internet-derived content, and is the trigger currently prevented?** A coarse component archetype can answer the first half. Current assessment, DAST, remediation, and retest evidence can support the second half for the trigger or input-control classes they actually cover.
4. **Can a reachable entry point supply local execution?** For an active AV:L finding, look within the same verified execution boundary—ordinarily a host and its kernel—for another active finding that is directly reachable and classified as a high-confidence chainable entry point. If one exists, promote the local finding to IRV and retain the upstream CVE and classification reason. Mere co-residency with an AV:N finding is not enough.

The paper calls the prevention fact in check 3 the P_0 assertion. If P_0 is current and in scope, a known downstream trigger can be classified NIRV. If the evidence is missing, stale, contradicted, or unrelated to the trigger, P_0 is zero and the finding remains IRV.

What counts as a high-confidence entry point

The strongest reason is authoritative or reproducible CVE-level evidence that exploitation supplies local code or command execution. When that evidence is absent, the operational policy uses narrow, governed signals rather than assuming that every severe network vulnerability is an entry point.

An AV:N CVE is classified high confidence when it maps to a strict execution-semantics seed: CWE-78, CWE-94, CWE-95, CWE-96, CWE-98, CWE-553, CWE-624, or CWE-917. A second rule requires two signals together: full vulnerable- system impact—C:H/I:H/A:H in CVSS v3 or its CVSS v4 equivalent—plus CWE-97 or CWE-494. The same full-impact combination may use CWE-1336 or CWE-829 only when governed evidence confirms that execution occurs in the affected server or runtime, not in a client.

Full impact by itself is only a possible entry point. So is an AV:N mapping to CWE-97, CWE-1336, CWE-494, CWE-829, or CWE-470 without the required corroboration. Full impact plus CWE-470

also remains possible. Possible candidates stay visible for filtering and review, but they do not automatically promote local findings.

These are versioned triage rules, not declarations that every matching CVE is proven remote code execution. A CWE describes a weakness family; it does not by itself establish the exact result of exploiting a particular CVE. Stronger vendor, CNA, exploit, Vulnrichment, or approved analyst evidence can replace the heuristic classification.

The evidence already exists, but it has limits

For services following the Continuous Monitoring Playbook, the FedRAMP Continuous Monitoring Playbook requires recurring web-application scanning, coverage of web interfaces or an approved sample, individual tracking of findings, and assessor-validated scanner configuration. Providers maintain that evidence and remediate findings. Assessors review its scope and integrity and verify provider-run scan results.

That division of labor matters. The provider produces and refreshes the evidence. The 3PAO collects or reviews it during assessment and validates the method. The model does not ask the assessor to invent a complete application data-flow graph or research every CVE on the provider's behalf.

A current, properly scoped DAST program is meaningful sanitation and validation evidence. It is also not universal proof. Automated testing can miss authenticated paths, unusual formats, new trigger techniques, and code that the test plan never exercises. The OWASP Web Security Testing Guide makes the same point: automated scanning belongs in a balanced testing program.

The practical line is straightforward:

- A clean, current scan with validated scope, relevant coverage, remediated findings, and successful retest can support P_0 for the tested class.
- A generic clean scan with unknown scope cannot.
- A new relevant finding, failed retest, material application change, uncovered interface, alternate path, or new trigger technique revokes or narrows P_0 .

Why reachability converges toward accessibility

The model makes three explicit triage presumptions.

First, a CVE whose indirect-trigger status is unknown is not automatically promoted across every component that ever handles internet-derived data. Doing that would make nearly every connected system IRV and erase the prioritization the field is meant to provide. The unknown status remains visible and is queued for risk-ranked enrichment.

Second, a current P_0 assertion is trusted within its validated scope until contrary evidence or a material change revokes it. This reuses the assessment and ConMon work providers already perform.

Third, an active AV:L finding is promoted only when a directly reachable, high-confidence entry point exists in the same verified execution boundary. Full impact alone and possible-only entry points do not cause automatic promotion. When the last qualifying entry point is resolved, loses direct exposure, or leaves the boundary, the chained promotion is withdrawn.

The resulting operational IRV list therefore converges toward the directly internet-accessible surface, plus known indirect triggers that are not covered by current prevention evidence, plus

boundary-scoped local findings promoted through high-confidence entry points. Reachability and accessibility do not mean the same thing. They produce similar operational lists because tested prevention closes many downstream trigger paths, while the exceptional content and chained paths are promoted explicitly.

That is a conditional convergence, not a promise that downstream content is always safe. Its residual risks are equally explicit: missing upstream trigger or exploit-outcome metadata, an imperfect entry-point heuristic, an incorrect execution-boundary assignment, and incomplete or stale prevention evidence.

Four examples

- **SQL construction behind an application tier.** Internet-derived values reach a private database. The validated DAST program covers SQL injection on the relevant interface, relevant findings were remediated and retested, and no bypass or material change is known. The current P_0 assertion makes the database finding NIRV. A new SQL-injection finding or uncovered interface resets P_0 , and the finding returns to IRV.
- **A logging-expression trigger.** Threat intelligence confirms that a crafted string can cross process boundaries and trigger the affected logging library. The component receives internet-derived log content. Ordinary SQL-injection and cross-site-scripting DAST does not cover this mechanism, so P_0 is zero and the finding is IRV.
- **An administrative service behind IAP or VPN.** A separate, bypass-free edge authorizes the connection before traffic reaches the backend and is the backend's only ingress path. That can remove the backend's direct internet path. A login implemented by the backend application does not do the same thing because the request has already reached the component; application authentication instead informs exploitability or an attested mitigation.
- **A network entry point chained to a local kernel finding.** A public web service has an active, directly reachable AV:N CVE. The CVE has full vulnerable-system impact and maps to CWE-97, so the combined-signal rule classifies it as a high-confidence entry point. An active AV:L kernel finding exists in the same host/kernel execution boundary. The kernel finding becomes IRV, keeps its AV:L vector, and records the web-service CVE as the reason. Full impact without the corroborating execution signal, an unexposed upstream finding, or a finding in another boundary does not cause that automatic promotion.

What happens when a new trigger appears

Log4Shell is the useful pattern. Before the trigger was understood, ordinary scanner output and web testing did not say that a string written to a log could activate a private library. Once the mechanism became known, the CVE-level indirect-trigger status could be promoted across every affected deployment archetype that received attacker-controlled log content.

That promotion is why trigger metadata belongs upstream. Research it once, publish it with provenance, and let every scanner and vulnerability-management platform reuse it. Exploit-outcome and execution-context metadata should work the same way for chainable entry points. Agentic analysis can propose missing profiles, but it should cite sources, preserve unknowns and possible-only statuses, and require stronger review before concluding that no indirect trigger or

execution outcome exists. An agent cannot recover a technical fact that no authoritative source disclosed.

When new intelligence shows that an existing prevention assertion does not cover the trigger, P_0 is revoked and the affected findings are re-evaluated. The control is not allowed to remain true merely because last month's scan was clean.

The optional future state is not today's requirement

A richer system could import or derive a deployment graph, record how content moves between components, attach transformations such as parameterized queries to each path, represent local execution boundaries and exploit outcomes, and bind tests to the deployed artifact. That would improve precision and make stronger IRV and NIRV claims portable between tools.

The industry does not yet have a broadly adopted way to combine those artifacts into an independently verifiable, vulnerability-specific reachability record that vulnerability-management platforms can consume. The full paper includes a non-normative equation showing how such a system could work. It is an interoperability direction, not a prerequisite for the current method and not a claim that providers must build a complete application-wide graph now.

A brief word about WAFs

FedRAMP allows a verified perimeter control that blocks the triggering payload before vulnerable processing to remove IRV status. The paper recommends a more durable operational posture for volatile WAF rules: keep the intrinsic reachability visible and attach a signed mitigation or VEX assertion describing the rule, its scope, and its evidence. If the rule is bypassed, disabled, or routed around, the record fails visibly instead of leaving behind a stale NIRV classification. KEV remediation dates continue to apply even when a vulnerability is fully mitigated.

The bottom line

1. Reachability is evaluated per finding, not assigned once to an asset.
2. Direct exposure can be automated from network, listener, protocol, and runtime evidence.
3. Exceptional indirect triggers should be enriched once per CVE and reused across the fleet; today's scanners rarely provide that metadata.
4. Active AV:L findings are promoted only through a directly reachable, high-confidence entry point in the same verified execution boundary, with the upstream CVE and reason preserved.
5. Existing assessment, DAST, remediation, and retest evidence supports a scoped, revocable prevention assertion without requiring a complete data-flow graph.
6. Under those declared presumptions, the operational IRV list converges toward direct accessibility plus known uncovered indirect triggers and narrowly governed chained-access cases.

This is a scalable triage model, not exhaustive exploit-path proof. It states its presumptions, preserves unknowns, and identifies the metadata needed for greater precision. *Read the full technical paper.*